Fog Removal Techniques from Images: A Comparative Review and Future Directions

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***Abstract*— Fog, haze and smoke are a big reason of road accidents. Fog reduces contrast level of the image that affects the visual quality of the image. In field of computer vision visual quality and visibility level of an image is affected by airlight and attenuation phenomena. Air particles, which present in atmosphere and affect the visibility level of any object, are called noise or unwanted signal between observer and object. For improving the visibility level of an image and reducing fog and noise various image enhancement methods are used. After enhancement is again restored the enhanced image by restoration methods. For improving the visibility level 4 major steps are used. First step is acquisition process of foggy images. Second is estimation process (estimate scattering phenomena, visibility level). Third is enhancement process (improve visibility level, reduce fog or noise level). Last step is restoration process (restore enhanced image).**

The main aim of this paper is to review state-of-art image enhancement and restoration methods for improving the quality and visibility level of an image which provide clear image in bad weather condition. We also compare prevalent approaches in this area through implementation of the methods keeping parameters common for critical analysis. In the end we provide the future scope for working directions in this area for the readers.

Keywords—***image acquisition, air light, attenuation, visibility enhancement, contrast enhancement, Fog removal, FFT.***

I. Introduction

Fog is a collection of water droplets or ice crystals draped in the air at or near the earth surface. Fog in form of cloud is known as stratus cloud. Fog is prominent from mist only by its density. Fog reduces visibility to less than 1km where as mist reduces visibility to no less then 1km.

A. Type of fog

Radiation fog, Ground fog, Advection fog, Evaporation fog, Arctic sea smoke, Precipitation fog, Upslope fog, Freezing fog, Frozen fog, Artificial fog are some of most recurring fog type .

TABLE 1: Visibility and weather Condition of Fog, Mist, Haze

	VISIBILITY	**WEATHER CONDITION**
FOG	Visibility less then 1km.	Cloudy
MIST	Visibility between 1 & 2 km.	Moist
HAZE	Visibility between 2 & 5km.	Dry

B. Effect of fog

Effect of fog mainly is caused by two scattering [1, 2] phenomenon-

1. Attenuation
2. Airlight

Attenuation –The light beam coming from a scene point gets attenuated because of scattering by atmospheric particles, called attenuation that decreases the contrast of the scene. Fog particles add into 3D GIS (Geographic information system) [3].

Airlight –The light coming from a source is scattered toward camera and give on to the shift in color, its called airlight. The fog effect is the function of the distance between scene point and viewer or camera. Hence, removal of fog requires the estimation of airlight map [4, 5].

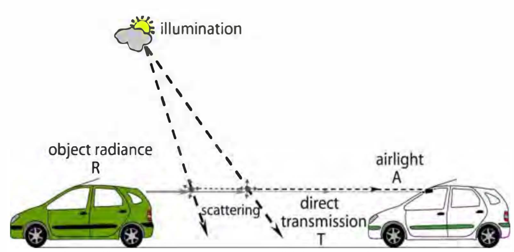


Fig 1. Scattering phenomena Airlight and Attenuation

The depth information of any input image is measured in term of airlight [6], transmission map [1], depth map or some time depth information is estimated with the use of scene properties. Initial work in fog removal is based on the contrast enhancement and restoration based approaches.

Under fog, haze and smoke weather conditions, the contrast and color characters of the images are degraded in a drastic manner. On Clear day images have high contrast compare than foggy images. Thus, a fog removal algorithm should improve the scene contrast [7].

TABLE 2: Properties of Particles

Condition	Particle Type	Radius (micrometer)
Air	Molecule	10^{-2}
Haze	Aerosol	$10^{-2.1}$
Fog	Water Droplet	10^{-10}
Cloud	Water Droplet	10^{-10}
Rain	Water Droplet	$10^{2} - 10^{4}$

Fog reduces visibility and contrast level of an image. To improve the quality of images various enhancement methods are used. A step by step image processing is applied over an image. Firstly acquire the image from real world and convert into system readable form, measurement of the effect of noise on the image. There are different types of noise which affects the image. Accordingly, image enhancement process for improving the quality of an image is needed then. After improving the quality of an image again restore that image. At present technology for fog removal are of two types-

- Fog correction
- Fog removal

The Fog correction is based on correction of contrast level. Color correction process is applied over HSV color space [1]. Color correction process generates transmission map and estimates atmospheric light resulting a defogging image. Further using color correction method enhanced video is created. Fog correction process improves the quality of foggy pixel but in fog removal process the fog level over an image is found out and removed. Figure 2. shows basic model for fog removal from images.

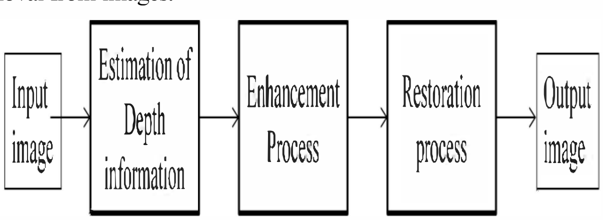


Fig. 2. Framework for fog removal

II. IMAGE ACQUISITION MODEL

When try to remove the fog for an image or video then first step is acquiring that input image or video. Image is acquired by camera. Videos can be considered as sequence of image frames. All the single image fog removal algorithms may be applied over the video frames also by regularly iterating them over each frame. Capturing of the image is done through CCD/CMOS sensor camera etc. Sensor plane are used for acquisition [1]. The irradiance received by cameras [2] when applies processing in real time system then use vertical camera [3] for capture image is also called on board vision. Authors in [3] used camera to convert 4D light field of a scene from 2D camera.

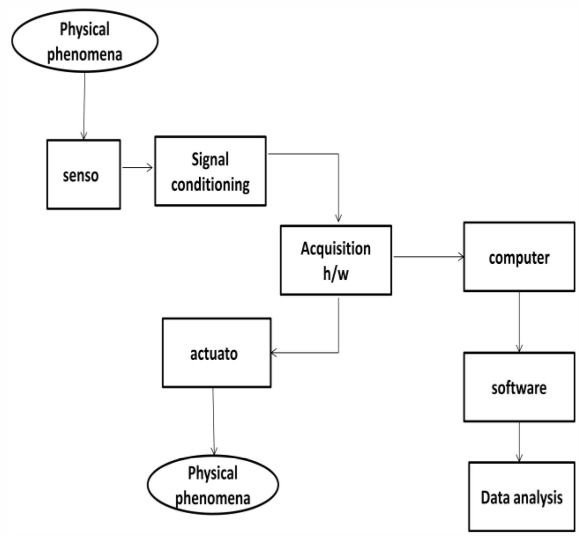


Fig. 3. Acquisition Model

Satellites are used [5] when working over optical wireless links. Image may also be captured using polarization filter, which works on different degree of polarization [7]. In physical model foggy image is describe as-

$$I(x) = J(x)\, \boldsymbol{t}(x) + A\, (\mathbf{1} - \boldsymbol{t}(x)) \quad \mathbf{(1)}$$

In equation (1), I is observed intensity, J is the scene Radiation, A is global atmospheric light, t is medium transmission describing the portion of light that is not scattered and reaches the camera. Authors [3] use in vehicle cameras and onboard vision. Onboard exteroceptive sensor used here to measure the proximity of any object with contact sensor and range sensor. Optical sensors are used over automotive vehicle [8]. For measuring the atmospheric visibility distance charge coupled camera used for acquisition process [9]. Transmissometer and scatterometer are used for measuring the visibility of light [10]. Advanced driver assistance system (ADAS) is used by [11] for acquiring the emotion features. Optical devices are used for capturing the image is most common [11, 12].

III. Estimation Based Model

After acquisition process, need to estimate the level of fog or noise in an image. Estimation of airlight, attenuation and depth map is basic factor which are calculated for enhancement process. Using equation (1) the airlight and balanced the pixel which have maximum brightness among all color channels is found out. In image which is effect by fog. K. He, J. Sun, and X. Tang [4] estimate the direct attenuation J(x) t(x) and air light A(1-t(x)) then estimation transmission for haze removal method over a single input image. For refining transmission map apply soft matting algorithm in same method. Dark prior channel is another factor which estimates for enhancing haze channel over an image. Dark prior channel is based on key approach - haze free outdoor image contain some local patch (non sky patches) which contain some pixels, which have very low intensities in at least one color channel. Using this prior estimate the thickness of haze and recover it and get haze free outdoor image. For calculating dark prior channel-

$$j^{dark}(x) = \min_{c\in\{r,g,b\}} \left(\min_{y\in\Omega(\mathbf{x})} (j^c(y))\right), \quad (2)$$

Equation (2) $j^{dark}(x)$ represent the dark channel of image x, If intensity value of $j^{dark}(x)$ is low or zero then haze free outdoor image is obtained. j^c Represent the color channel and $\Omega(\mathbf{x})$ represents local non sky patches for input haze image.

During use of dark prior channel for haze removal transmission map is estimated. Assume that local patches are constant for dark prior channel than patch's transmission is $\check{t}(x)$ and perform the operation in local patches is-

$$\min_{y\in\Omega(\mathbf{x})}(I^c(y)) = \tilde{t}(x)\min_{y\in\Omega(\mathbf{x})}(J^c(y)) + (1-\tilde{t}(x))A^c \quad (3)$$

In equation (3) I^c represent the observed intensity of color channel and A represent airlight which describe in equation (1). This min operation is perform over 3 color channel independently then equation is

$$\min_{y\in\Omega(\mathbf{x})}\left(\frac{I^c(y)}{A^c}\right) = \tilde{t}(x)\min_{y\in\Omega(\mathbf{x})}\left(\frac{J^c(y)}{A^c}\right) + (1-\tilde{t}(x)) \quad (4)$$

Then, min operation among three color channel on equation (4) and obtain –

$$\min_c\left(\min_{y\in\Omega(\mathbf{x})}\left(\frac{I^c(y)}{A^c}\right)\right) = \tilde{t}(x)\min_c\left(\min_{y\in\Omega(\mathbf{x})}\left(\frac{J^c(y)}{A^c}\right)\right) + (1-\tilde{t}(x) \quad (5)$$

In dark prior channel, j^{dark} of haze free radiance j tends to zero.

$$j^{dark}(x) = \min_c\left(\min_{y\in\Omega(\mathbf{x})}(j^c(y)\right) = 0 \quad (6)$$

As, airligtht of color channel is always positive then

$$\min_c\left(\min\left(\frac{J^c(y)}{A^c}\right)\right) = 0 \quad (7)$$

Equation (7) put in eq. (5) and estimate transmission map directely.

$$\tilde{t}(x) = 1 - \min_c\left(\min\left(\frac{J^c(y)}{A^c}\right)\right) \quad (8)$$

Transmission map in used in many other algorithms rather than dark prior channel. Depth map is also estimated with the help of transmission map function.

IV. Enhancement Based Model

After estimating the factor of noise in term of airlight, attenuation, transmission map and depth map be applied enhancement process for improving the image of contrast level and visibility quality of an image.

Image enhancement process is divided into 2 parts, based on pixels. One is spatial domain and another is frequency domain.

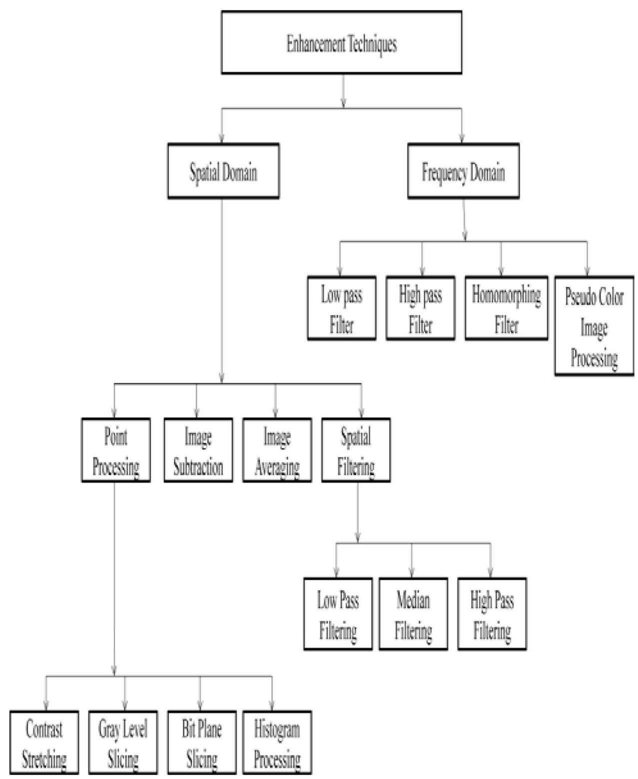


Fig. 4.Classification of Enhancement Techniques

In spatial domain, process is applied over a pixel of input image. In image processing spatial domain function is-

$$g(x, y) = T[f(x, y)] \quad (9)$$

In equation (9) f(x, y) is the input image and g(x, y) is the output image and T is the operation that over pixels of image f(x, y).

In frequency domain used frequency domain filters which is based on Fourier Transformation of an image for enhancement process.

Contrast enhancement techniques are divided into two classes.

A. Non Model Based Enhancement Techniques

In non model based enhancement techniques enhancement process is applied over acquired input image or video. In this process no need to collect extra information about input images. Histogram equalization is the most common method is based on non model based enhancement. Many other enhancement based method like unsharp masking [13], wavelet [13], retinex theory [13], for improving the contrast, brightness of any image. Using retinex algorithm find the balance between the human vision and machine vision, the retinex algorithm classified by 3 ways- 1) Single – scale retinex (SSR) 2) Multi – scale retinex (MSR) 3) Multi scale retinex with color restoration (MSRCR). Histogram equalization is a contrast enhancement technique that adjusts pixels intensities in order to obtain new enhanced image [14] with usually increased local contrast.

Histogram equalization are divided into different type-

1. Histogram equalization (HE)
2. Adaptive histogram equalization (AHE)
3. Contrast limited adaptive histogram equalization (CLAHE)
4. Brightness preserving bi histogram equalization (BBHE)
5. Quantized bi histogram equalization (QBHE)
6. Dual sub image histogram equalization (DSIHE)
7. Minimum means brightness error Bi-histogram equalization (MMBEBHE)
8. Recursive mean separate histogram equalization (RMSHE)
9. Brightness preserving dynamic histogram equalization (BPDHE)
10. Weighting mean separated sub histogram equalization (WMSH).

1. HISTOGRAM EQUALIZATION (HE)

Histogram equalization is a method of improving the contrast globally. This process is the adjustment of intensity which is globally distributed across the image. If we consider Any gray scale image (x), n_i be the number of occurrence of gray level i and a probability function of occurrence of a pixel of level i in image (x) is

$$p_x(i) = p(x = i) = \frac{n_i}{n}, 0 \leq i < L \quad (10)$$

In paper [15] histogram equalization method are used to adjust intensity values separately for background and foreground images after that image are fused into new frame. 'imhist' function used for equalization in Matlab.

Fig. 5. (a)Original image

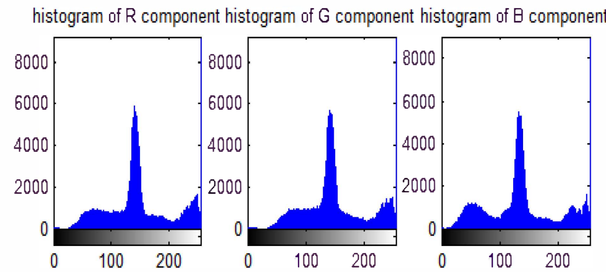


Fig. 5(b) Histogram of Original foggy image

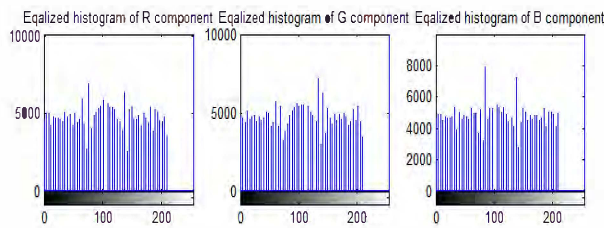


Fig. 5(c) Equalized histogram for RGB layer

Fig. 5(d) after apply histogram equalization method restored image

2. ADAPTIVE HISTROGRAM EQUALIZATION (AHE)

Adaptive Histogram equalization method is a modified part of histogram equalization method. In this method enhancement process are applied over a specific region of any image and adjust contrast according to their neighbor pixels. Inhye Yoon used AHE for enhancement in HSV color space [16]. 'adapthisteq' function used in Matlab for equalization of intensity according to neighbor pixels. This method is used only for homogeneous fog correction. Gyu-Hee Park, Hwa-Hyun Cho [17]Dynamic range separation Histogram equalization (DRSHE) method for enhancement. In this method histogram divided into some predefined parts. Then adjust the intensity of that that part and uniformly distribute into gray scale image.

3. CONTRAST LIMITED ADAPTIVE HISTROGRAM EQUALIZATION (CLAHE)

Contrast limited adaptive equalization is a modified part of adaptive histogram equalization. In this method enhancement function is applied over all neighborhood pixels and transformation function is derived. This is differing from AHE because of its contrast limiting. Zhiyuan Xu, Xiaoming Liu, Xiaonan Chen [18] used CLAHE method for enhancement process, in this method used maximum value to clip the histogram and redistribute in gray level image. Algorithm applied separately for background and forground [18] and limit the noise, enhance the contrast. Distribution parameter are used for shape of histogram equalization graph, in paper [18] 'Rayleigh' distribution parameter are used for bell shaped histogram. CLAHE applied over both type of images gray scale and colored. 'cliplimit' function is used and apply limit over a noisy image. LAB color space is used for RGB true color images.

4. BRIGHTNESS PRESERVING BI HISTOGRAM EQUALIZATION (BBHE)

After histogram equalization process brightness of an image is affected which create a problem in some electronic products like telivision. For removing this draw back BBHE is used. In this method [7] input image is divided into two sub images and histogram equalization process is applied separately over two sub images. After this method is applied the output image which preserving mean brightness then normal histogram is used to get naturally improved enhancement image that can be utilized for electronic products.

5. QUANTIZED BI HISTOGRAM EQUALIZATION (QBHE)

Histogram equalization changes the mean brightness of original image for histogram flattening. This is the drawback of normal histogram method when original brightness is required. For removing this drawback another version of BBHE is used. QBHE this is based on cumulative density function of a quantized image [19] and average intensity value of their separating points. In paper [19] an algorithm for simple hardware structure and used QBHE method provided enhanced image which is useful for consumer electronics.

6. DUAL SUB IMAGE HISTOGRAM EQUALIZATION (DSIHE)

Dual sub image histogram equalization uses median intensity value of their separating points.

7. MINIMUM MEANS BRIGHTNESS ERROR BI-HISTOGRAM EQUALIZATION (MMBEBHE)

Brightness preserving histogram equalization removes the drawback of normal histogram equalization but in this method also have some drawback. In some cases required high degree of preservation which is not handled by BBHE, for removing minimum mean brightness error used MMBEBHE [20]. This algorithm provides maximum brightness preservation. In this method input image is divided into sub images in basis of threshold values, which would yield minimum absolute mean brightness error (AMBE). Result of this approach present the sample enhanced image with very low, high, medium mean brightness.

8. RECURSIVE MEAN SEPARATE HISTOGRAM EQUALIZATION (RMSHE)

Recursive mean separate histogram equalization is improved version of BBHE. This method provides better and scalable brightness preservation result compare than BBHE method. BBHE separates the input image on the basis of its mean before equalizing them independently but in RMSHE method separation is performed recursively; separate each new histogram based respective mean. Using recursive mean increases the brightness of input image and enhanced image is obtained [21].

9. BRIGHTNESS PRESERVING DYNAMIC HISTOGRAM EQUALIZATION (BPDHE)

This approach is extension of histogram equalization. In this process [22] produced the output value with mean intensity which is equal to the input image mean intensity which overcomes the problem of maintaining the mean brightness of an image. Smooth the input image histogram and then partition the smooth histogram based on its local maximums. New dynamic range provides in each separated histogram and applied histogram equalization method independently, which is based on new dynamic range. Last step of this algorithm is normalization process of output image to the input mean brightness. After this process get the enhanced input image.

10. WEIGHTING MEAN SEPARATED SUB HISTOGRAM EQUALIZATION (WMSH).

Histogram equalization improved the contrast enhancement but effect the visual property of image this is the major drawback of HE. For removing this problem the approach of weighting mean separation is used [23]. Precise histogram equalization is used in this approach with piecewise transformed function. Using this method produced enhanced output image with original visual properties.

B. Model Based Enhancement Techniques

In model based method physical models are used to forecast the pattern of image degradation information. It requires extra information about the imaging environment.

De-fog

Blur is factor that is responsible for reducing the visibility of images. De-fog method for improving the quality of image using depth of the image using blur estimation.

De-haze

Haze is term used in image analysis, which is a set of atmospheric effect that reduces the contrast of an image. For removing haze a simple and effective method used [12]. This method does not require user interaction; it can be used in practical real time application. This method is also useful when it comes to an offline system.

V. Restoration Model

After enhancement and estimation process again restores the fog free image. For single image haze removal [24, 4]recovering the scene radiance for restoration. J(x) is scene radiance is recovered by

$$J(x) = \max(t(x), t0) + A \quad (11)$$

In paper [10, 11] Koshmieder's law are used. Inverse of Koshmider's law are also used for restoration process [24]. Gamma correction and tone mapping are used before restoration of image for improving the quality of enhanced image.

(a)

(b)

Fig.6. (a) Input image (b) Restored image

Multiple restoration method are used for multiple image and single Images-

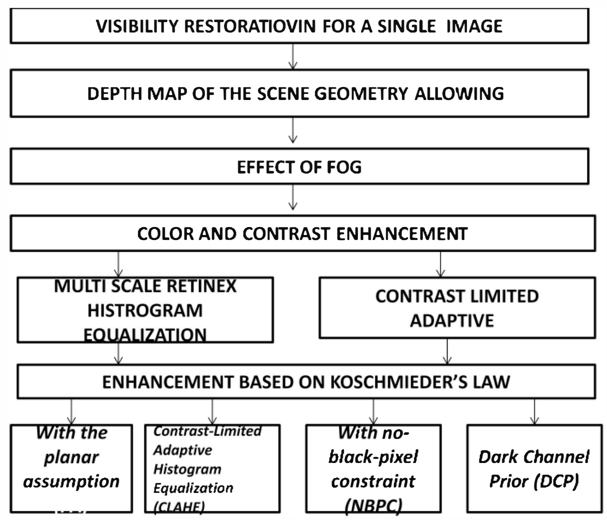
- Knowledge of scene depth method use for multiple input images but only gray input images [27].
- Light scattering by atmospheric particles in partially polarized method for color and gray both images [28].
- Uniform bad weather condition method for color and gray both type of input images [40].

For single input images-

- Interactive
- Airlight is contrast throughout theim – age for color and gray images [44].
- Cost function based on human visual model.
- Shading and transmission function are locally uncorrelated for colored single image only.
- Method based on local contrast and spatial regularization for color or grays both images [29].
- Dark channel prior for single input images [4].
- Airlight assumes as contrast percentage for color and gray level images.
- Restoration based on graph based image segmentation.
- Anisotropic diffusion for color or gray single input image [30].

VI. Compression Based Model

Compression model is applied between two frames [2] which are consecutive in term of color, distortion and HSV color channel. Video compression is used in paper [28]. A comparison is based on visibility level of any image. In paper [21.2] the results show the comparisons between different the different enhancement technique which used Koshmider law for restoration process of enhanced images [4, 31].



VII. Discussion and Comparison

Nicolas Hautiere and Jean-Philippe tarel [32] proposed a method for estimation of visible edge in foggy image and visible edges of same image after contrast restoration process. Sobel gradient filter are used for finding the gradient of both images which use 3*3 window size filters over an image and find out the visible edges. This process is applied over contrast restored image at 5%. Nicolas proposed algorithm for finding out rate of new visible edges.

$$e = \frac{n_r - n_o}{n_o} \quad (12)$$

(a)

(b)
Fig.7. (a) Original image (b) Restored image

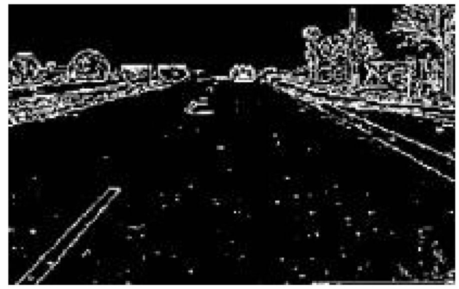
In equation 12 n_o and n_r are cardinal number of visible edges set in image (original image) and I_r (contrast restored image) Respectively and latter set is represent by ρ_r and calculate the r, which represent the quality of restoration by method Proposed by Nicolas[23].

$$r = exp \frac{1}{n_r} \sum_{p \in r} log r_i \qquad (13)$$

In Eq. 13 n_s is number of saturated pixels (black and white) which calculate contrast restoration and normalized the size value of image represent by σ in this equation 4.

$$\sigma = \frac{n_s}{dim_x * dim_y}$$

(a)



(b)

Fig. 8 (a) Visible edges in original (b) visible edges in restored image

Inhye yoon, Seonyung kim and D.kim used adaptive defogging method in paper [1]. They present an algorithm for correction of color in HSV color space for defogging. For this process image is acquired by video then create a modified transmission map and estimate the atmospheric light for contrast enhancement and estimation process. After getting restored image we calculate the same quality parameter which calculate in paper [32] and find out the comparatively better result. In paper [1] generate the transmission map then find out how much light reflect by object and reaches the camera. Consecutive frames are used for recovery of de-foggy frames.

Fig. 9 (a) original image (b) restored image

Visible edges rate is increase in methodology used by I. yoon paper because in this algorithm contrast restoration process is applied over each layer and pixel which produce the more visible edges as compare to previous method proposed by Nicolas Hautiere [32]. Quality of restoration which is denoted by r is also increased by 0.2% compare then previous method [15] So, the algorithm proposed by [1] in is better than algorithm proposed by [15].

(a)

(b)

Fig. 10(a) visible edges in original image (b) visible edge in restored image

TABLE 3: QUANTITATIVE EVALUATION OF TWO ALGORITHMS

Algorithms	**Visible edges in original image**	**Visible edges In restored image**	**e**	**σ**	**r**
Adaptive defogging [1]	39904	96394	1.4156	0.00041%	2.7148
Gradient Ratioing At visible edges [23]	14184	22635	0.59581	0%	2.5387

In paper [45] contrast enhancement techniques use the visible edges ration that compute for each pixel of visible objet and this algorithm applied only daytime fog image. In this approach each pixel having local contrast 5% only and process based on segmentation algorithm. Four descriptors are used for enhancements image-visibility level enhancement for each image before and after contrast enhancement. Estimate the rate of new visible edges.

VIII. CONCLUSION AND FUTURE WORK

Many algorithms is developed for improving the visibility quality of an image in spatial domain but if these methods are applied in frequency domain then it produce better result and reduce the time to produce the output. When we want to convert the spatial domain input into frequency domain then use Fast Fourier transforms (FFT) and Discrete Fourier transforms (DFT). When we get the frequency of noise then we use filtering methods and reduce the noise frequency and produced is enhanced output improved image.

$$G(k,l) = F(k,l)H(k,l) \tag{14}$$

Equation (14) represent the noise image filtering in frequency domain F(k.l) is input image in Fourier domain and H(k,l) is filter function applied over noisy image. G(k,l) is enhanced output image.

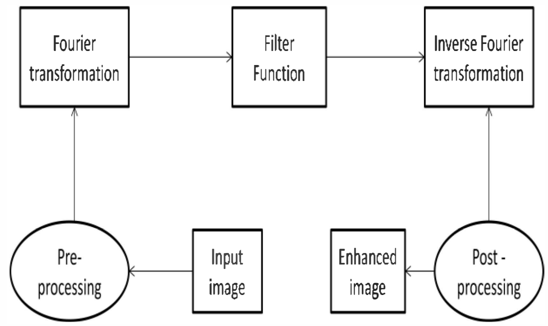


Fig.11. Enhancement process in frequency domain

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